

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int n;
4      scanf("%d", &n);
5      int a[n];
6      for (int i = 0; i < n; i++)
7          scanf("%d", &a[i]);
8      for (int i = 0; i < n; i++) {
9          for (int j = i + 1; j < n; j++) {
10             if (a[i] == a[j]) {
11                 printf("%d", a[i]);
12                 return 0;
13             }
14         }
15     }
16     return 0;
17 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

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Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57
3 10 17 57	
6	
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int T, n, m;
4      scanf("%d", &T);
5      while (T--) {
6          scanf("%d", &n);
7          int a[n];
8          for(int i=0;i<n;i++) scanf("%d",&a[i]);
9          scanf("%d", &m);
10         int b[m];
11         for(int i=0;i<m;i++) scanf("%d",&b[i]);
12         int i=0,j=0;
13         while(i<n && j<m) {
14             if(a[i]==b[j]) {

```

```
15         printf("%d ",a[i]); i++; j++;
16     }
17     else if(a[i]<b[j]) i++;
18     else j++;
19 }
20 printf("\n");
21 }
22 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

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3 10 17 57
6 2 7 10 15 57 246
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Output:

```
10 57
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Input:

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1
6 1 2 3 4 5 6
2 1 6
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Output:

```
1 6
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For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

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1 #include <stdio.h>
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3 int main() {
4     int T, n, m;
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7     while (T--) {
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12        scanf("%d", &m);
13        int b[m];
14        for (int i=0;i<m;i++) scanf("%d",&b[i]);
15    }
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15
16     int i=0,j=0;
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18         if(a[i]==b[j]) {
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25 }
26 return 0;
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Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int n, k;
4      scanf("%d", &n);
5      int a[n];
6      for(int i=0; i<n; i++)
7          scanf("%d", &a[i]);
8      scanf("%d", &k);
9
10     for(int i=0; i<n; i++) {
11         for(int j=i+1; j<n; j++) {
12             if(a[j] - a[i] == k) {
13                 printf("1");
14                 return 0;
15             }
16         }
17     }
18     printf("0");
19     return 0;
20 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

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